

17/01/2020

From Brain Signal to Language [Paul Smolensky]

Paul Smolensky

What we are at:

→ New kind of existence to co-exist with

→ "Final Revision" story to Claude

↳ NN, "neurons connected with each other" "Connectionism"

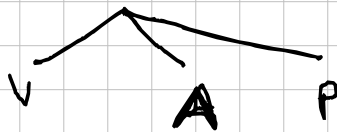
SYMBOLS: INPUT & OUTPUT → symbols, where
↳ recurrent activation vector

↳ How does it look from the p.o.v of COGNITIVE SCIENCE?

COGNITION $\approx$ COMPUTATION

A Pendulum...

Discrete symbolic structures (1955)
(manipulated by rewrite rules)



Numerical Vectors (1985)

PDP generation

Discrete symbol structures 1985
A = assigned probabilities

Numerical Vectors

Deep Learning generation (2005)

What kind of computational system is cognition?

↓
symbolic computation to understand models

↘ Neural Comp, LLM

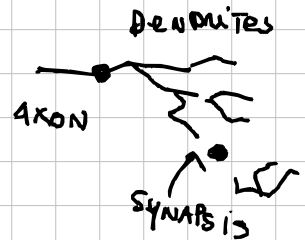
What is The secret of human intelligence?

PROGRAM: MIND

- REASONING
- MATH
- NLP

HARDWARE: BRAIN

- General properties
- Detailed properties



Discovering The implicit symbolic structure of LLM

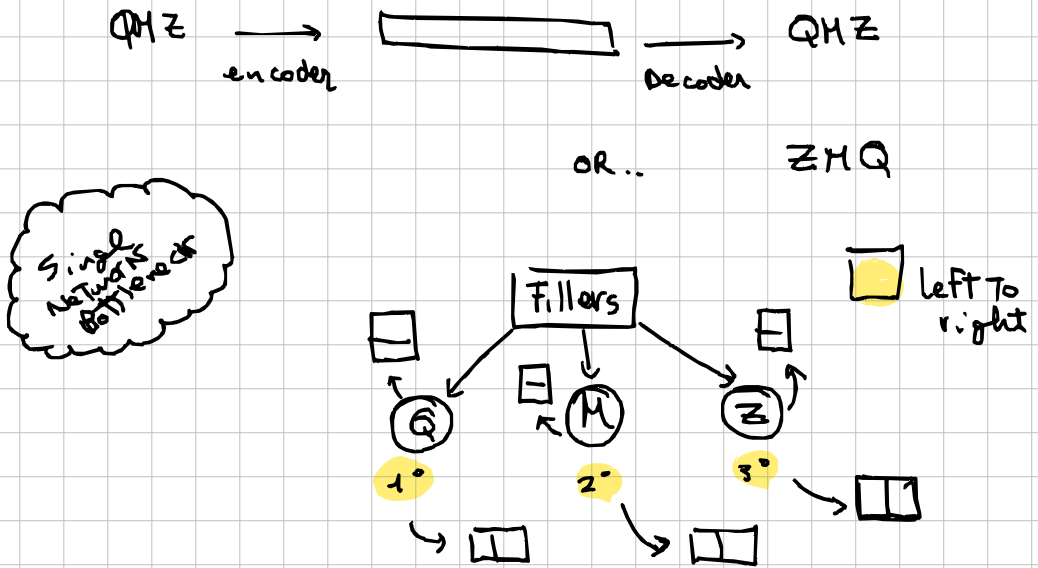
(Tom McCoy)

Dog chase cats
↑
Cats chase Dog

1 0 1 0 0 1

1 0 1 0 0 1

HP: They encode Tensor Product Representation



Tensor products of F and $[]$ $\rightarrow$ LINEAR MAP

NN are implicit TPRrepresenters

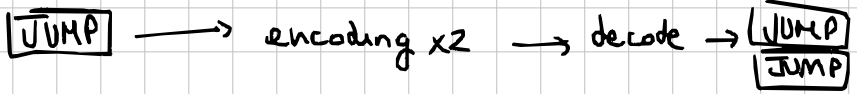
We Feed The TPRs To a decoder To check

MLP
RNN
Transformer

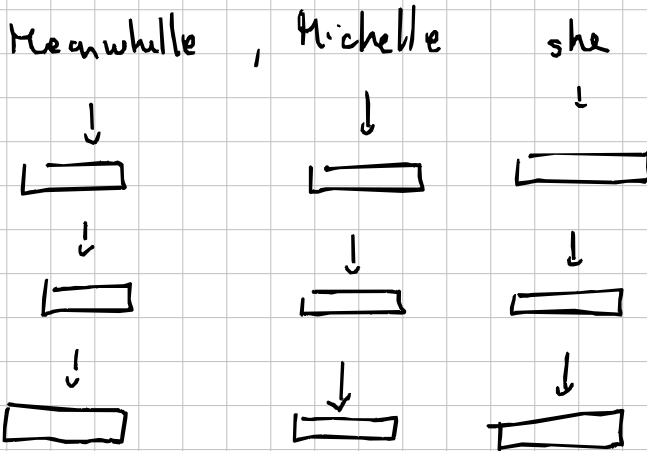
} Turns out BIDIRECTIONAL WORK 100% $\forall$ architecture

UNcovering TPR in Embed Representation, (Tom McCoy)

Jump twice

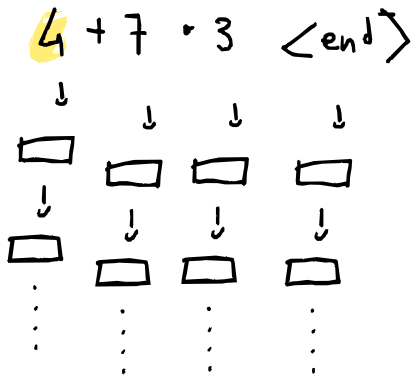


Prompting Claude



D. TPR appear in LLM as well?

- Arithmetic
- Logic
- Linguistic Transformations
- ...



Sequential
SYNTACTIC

We construct
hypothesis on
STRUCTURE

(Position, Pos ...)

25

TPR in GA-OSS

Summe performances
on these tasks

Can we do causal intervention? ✓

The clever doctor helps the lawyer

- clever: subject-adjective + clever: object-adjective

TPR explains other interpretability methods ✓

Also TPR connect

NN theory
(BRAIN)

→

SYMBOLIC
RULE THEORY
(MIND)

18/08/26

TPR enable complex processing of discrete language structures

1) How To explain human cognition?

Cognition is computation

How?

- Explicitly encodes and applies symbolic rules
Symbolic rules can not cover everything

↳ Symbolic residue

Ex. Next-word prediction vs Subj=verb dyademen

Cognitive Architecture

Foundation : Cognition in computation

What is computation?

□ Cognitive Functions	{	Pixels
		Sound Stream
	{	word string
HIGHER		Lexical Form

Computational Architecture : primitive + combinations

→ Pure Symbolic Architecture (PSA)

- Data: strings, binary trees

- Operations

Primitives

- Concatenate = cons
- First member = ex 0/1

Combinators

- Composition $f(x) = g(h(x))$
- Conditional branching

We have Discrete data (is or is not X)

EX. f passive → LF

Few leaders are admired by Donald Trump

The Connectionist hypothesis

FDP COMPUTATION

- Representation : Distributed activation vectors
- Primitive operations
 - Multiplication
 - Summation
 - Non-linear transfer function

$$a_\alpha = f(\sum_p w_{\alpha p} a_p) = F([W a]_\alpha)$$

- Combinations : parallelism

→ Data and operations of connectionism do not promise such capability

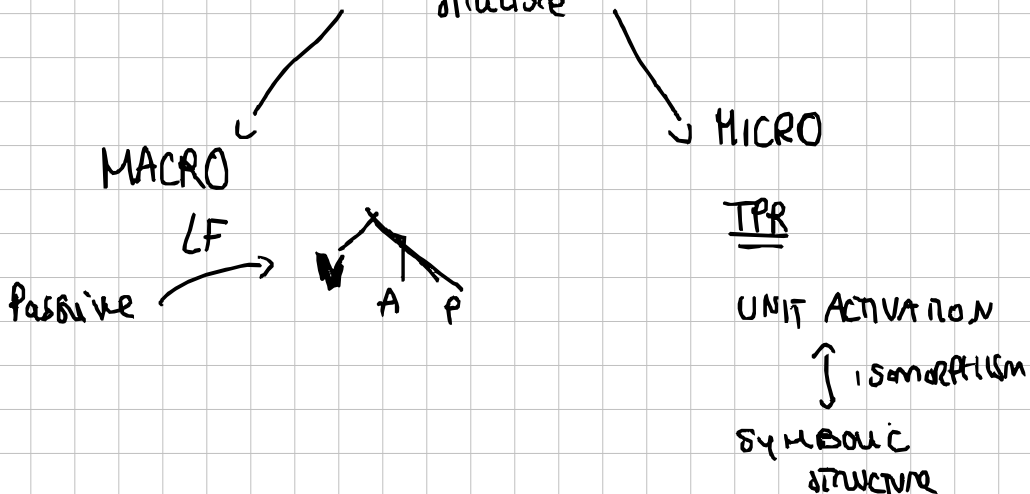
Trying to solve this problem: NECST

The NECST Hypothesis

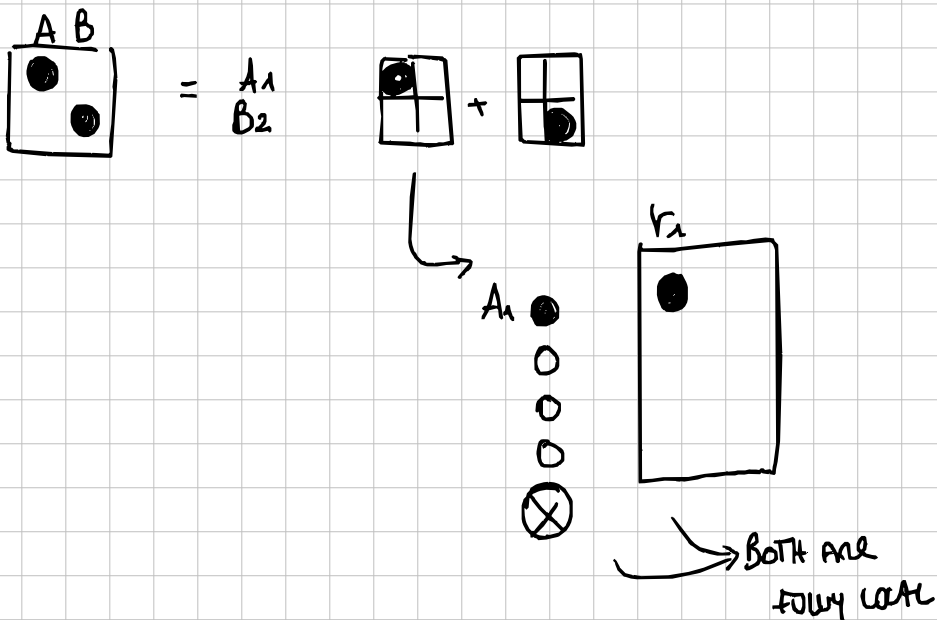
- In the core of HD domain, representation and function are well approximated by symbolic
- Connectionist hyp is correct
- Cognitive theory must reduce to PDP computation

NECST Isomorphism

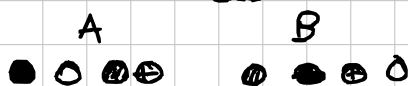
NECST Hypothesis: distinguish micro to macro structure



Fully local representation of letter strings



Semi-local

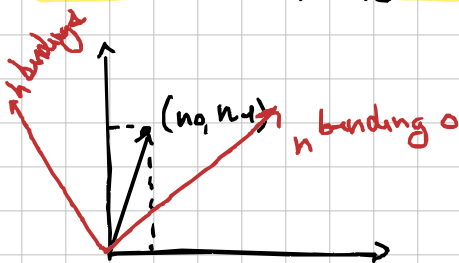


Fully distributed $AB = A \otimes r_1 + B \otimes r_2$
 $= \sum_{x \in F} x \otimes r_x$

no linear map W

- proper TFR: vectors for all fillers and all roles are linearly independent
- Perfect unbinding by inner product $\rightarrow$ can get fillers

From neural to conceptual coordinates

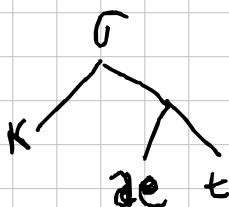


→ Map from activations to neuron to activations of concepts

Tensor product of Atoms

$$= \sum_k \text{internal state} \otimes \text{vector for orbital of electron } k$$

[TPR exhibit error analogues with King-men: queen-woman]



MACRO STRUCTURE

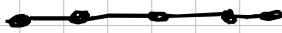
σ/r_3

k/r_0

ae/r_{01}

t/r_{11}

MICRO



$$\sum_i f_i \otimes r_i$$

Filler $\otimes$ role = filler/role bind a filler to a role
 Isomorphism micro $\leftrightarrow$ macrostructure

Recursion $r_0 \otimes r_x = v_{\text{left-child}}$
 $r_x \otimes r_x = v_{\text{right-child}}$

TYPE OF ROLES IN TPRS

Define: Structural Type } role }
 Structural token } filler/roles }

<u>EX.</u>	<u>EXAMPLE</u>	<u>CONTEXTUAL</u>	<u>POSITIONAL ROLE</u>
	K runs	K / runner $K \otimes \text{run}$	K / agent, run / V $K \otimes \text{agent} + \text{run} \otimes V$
	J loves K	$J \otimes \text{loves} \otimes K$	$J \otimes \text{agent} + \text{love} \otimes V$ + $K \otimes \text{patient}$
	[BC]	$B \otimes C$	$B \otimes \text{ro} + C \otimes v_1$
	[A[BC]]	$A \otimes (B \otimes C)$	

Computations over TreePR

Hand Picked

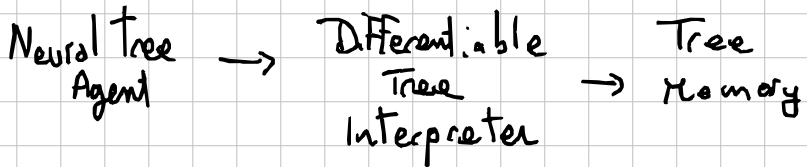
Can we do real symbolic processing with TreePR?

Semantics: β Reduction

Syntax: Tree Adjoining

Learned

Differentiable Tree Machine



Neural Tree Agent

③ NECS-Theory: discrete compositionality and grammar

General Symbolic Structures

[STRINGS ✓]
[TREES ✓]

What is a symbolic structure?
Has a number of ROLES, each role containing a SYMBOL

$\{R_1: S_1, R_2: S_2, \dots\}$

What primitives to embed in the DL model
Symbolic structures??

TPAs neurally embed the primitives of symbolic computations

- 1) binding
- 2) operation: binding, aggregation, unbinding
- 3) Fillers contain roles
- 4) Fillers can be structures
- 5) Roles can be recursive
- 6) Roles and fillers are mapped independently

→ Compositionality challenge for Connectionism

↳ to develop an effective micro-level formalism for compositionality that resides in vector spaces

|| NNV don't intuitively understand that the world is compositional!

→ encodings that are

- Compositional
- Continuous

"FRIEND" in French

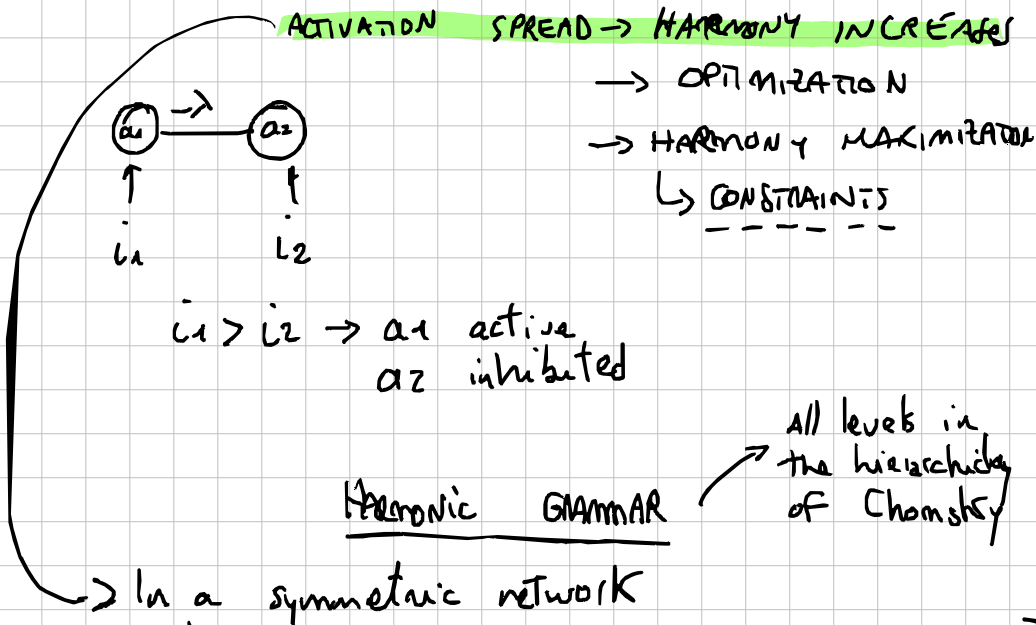
$[0.37 + 0.8n + 0.3t]ami$

Ex. "LOCKABLE"

LOCK [, -] & ABLE [-,]
LOCK: L ABLE: R

ACTIVATIONS - DISCRETE COMPOSITIONALITY and GRAMMAR

$$\frac{da_1}{dt} = -a_1 + i_1 - \lambda a_2$$



$$\frac{dH}{dt} \propto \sum_i \frac{da_i}{dt} [i_i] \geq 0$$

$$\left[\text{sign} \frac{da_i}{dt} = \text{sign } i_i \right]$$

What does $H_{\max}$ with TPR?

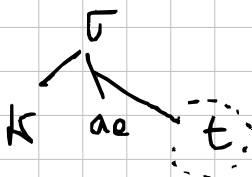
$\rightarrow$ MICRO-LEVEL

$\hookrightarrow$ MACRO-LEVEL

$$H = \sum_{\alpha\beta} a_{\alpha} W_{\alpha\beta} a_{\beta}$$

Constraints

No Coda $H < 0$
[Hawai, French, English]



Violation

Grammar of symbol structures

wellformedness $\uparrow$ optimization $\uparrow$ goodness
Neural Network Computation

(HARMONIC GRAMMAR $\rightarrow$ OPTIMALITY THEORY)

Discrete Harmonic Grammar : Discrete Input & Output

I want to express <RAINS>

	<u>SUBJECT</u>	<u>FULL INTERP</u>	
rains	*		-3
it rains		*	-2
it rains it		**	-4
rains it	*	*	-5



Dynamics = SGD + quant. straight

$\rightarrow$ CONSTRAINTS (different in my language)



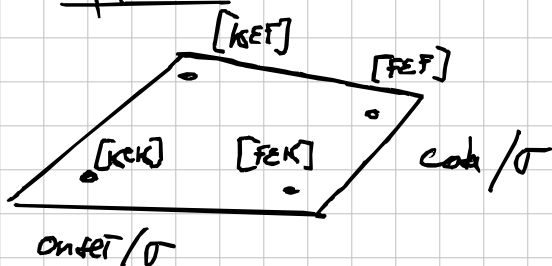
|| In the limit $t \rightarrow \infty$
 $T \rightarrow \infty$
|| TPR optimize Harmony

[Incremental sentence processing]

NON DISCRETE STRUCTURE OF COGNITION

Gradient (vector space structure) Continuous

Ex. Syllables



Phonology not
phonetics

$$\begin{matrix} \nearrow & \nwarrow \\ 0.7A & 0.4A \\ +0.2B & 0.9C \end{matrix}$$

$$0.7A \otimes r_{left} + 0.4 \otimes r_{right} \\ + 0.2B \otimes r_{left} + 0.9C \otimes r_{right}$$

3 Repr of a sentence $\begin{cases} \rightarrow \text{Phonological} \\ \rightarrow \text{Logical} \\ \rightarrow \text{Syntax Structure} \end{cases}$

What did Kay V?

Fix. LF: $Q_{what} . V(KAY, X_{what})$

↓ optimize
SS

SS: What Kay V

↓ optimize
PF

Gradient: REPRESENTATIONS IN GRAMMAR: LIASON -

IF we give up on DISCRETE Representations
Language may be better.

• /peti:t + ami/ & peti + tami (syllabic)
(orthography)

2 COMPETING
DISCRETE ANALYSIS

Structural blend hypothesis → LLM have
continuous

20/08/2026

LLMs: Generative AI & Generative Linguistics

"Do the syntactic abilities of generative
AI systems falsify fundamental principles of
Generative Linguistic Theory?"

① Computability → NL syntax has been
understood as abstract
symbolic computation

PRODUCTION SYSTEMS: has an ACTION
CONDITION
Computation plays out in Working
Memory

LLM

if The prompt displays a pattern, The LLM reproduces it → TEMPERATE GENERATION
(In context learning)

ex. Pure meaningless symbols → to study LLM ability to generate syntax
↳ good for gpt-4

Generation MUST be driven by abstract symbolic computation

How can some LLM solve This?

→ Do They have grammar rules hiding inside?
→ Could they? → at least TPR

About Attention ...

A) query-key can implement the CONDITION of a symbolic production

B) The Value can implement ACTION
→ states but looping

TRANSFORMER PRODUCTION FRAMEWORK (TPF)
(doesn't learn from data!)

Functional level: Discrete Function definition

Algorithmic: Symbolic Languages

Implementation: A numerical NN Transf implementing discrete repr. processes

→ Proves that LLM can be Tuning Universal symbolic Abstraction

Does intelligence arise from PRODUCTION ?
SYSTEMS :

→ Decomposing Representation Space into
Interpretable Subspaces with Unsupervised Learning

② LEARNABILITY : limited Knowledge of Rules of WM
→ NTP is enough to learn Syntax

→ From programming weights to learning them